

ACML101
Physical Chemistry Tutorial II

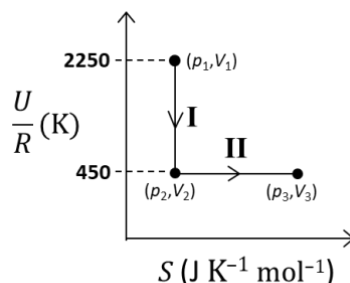
Instructor
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Q1. 1.00 mole of a monoatomic ideal gas at 298 K is expanded from 10 m^3 to 20 m^3 isothermally and reversibly. Calculate ΔS and ΔS_{surr} and ΔH in the process.

Ans: $5.76 \text{ J K}^{-1} \text{ mol}^{-1}$, $-5.76 \text{ J K}^{-1} \text{ mol}^{-1}$, 0

Q2.

Q.17 One mole of an ideal gas at 900 K, undergoes two reversible processes, **I** followed by **II**, as shown below. If the work done by the gas in the two processes are same, the value of $\ln \frac{V_3}{V_2}$ is ____.



(U : internal energy, S : entropy, p : pressure, V : volume, R : gas constant)

(Given: molar heat capacity at constant volume, $C_{V,m}$ of the gas is $\frac{5}{2}R$)

Ans: 10

Q3. 1.00 mole of a monoatomic ideal gas at 298 K is expanded from 10 m^3 to 20 m^3 isothermally against a constant external pressure of 10^5 Pa . Calculate ΔS and ΔS_{surr} , and ΔH in the process.

Ans: $5.76 \text{ J K}^{-1} \text{ mol}^{-1}$, $-3355.7 \text{ J K}^{-1} \text{ mol}^{-1}$, 0

Q4. 1.00 mole of a monoatomic ideal gas at 298 K is expanded from 10 m^3 to 20 m^3 adiabatically and reversibly. Calculate ΔS and ΔS_{surr} , ΔH and ΔT in the process.

Ans: 0, 0, $-1371.8 \text{ J mol}^{-1}$, -110 K

Q5. ΔH of vaporization of water is 539.4 cal/g at the normal boiling point. (a) Many bacteria can survive at 100°C by forming spores. Hence autoclaves used to sterilize medical and laboratory instruments are pressurized to raise the boiling point of water to 120°C . What will be the pressure needed to increase the boiling point of water to 120°C .

Ans: 1.94 atm

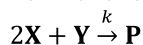
Q6. On the sea bottom at the Galapagos Rift, water heated to 350 °C gushes out of hydrothermal vents at a depth of 3000 m. Will this water boil or remain liquid at this depth ? The vapor pressure of water is 163 atm at 350 °C.

Q7. Consider a chemical reaction in which peroxydisulphate ions are used to oxidize iodide ions to triiodide. If rate of disappearance of peroxydisulphate ions is $1.37 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$, calculate the rate with which iodide ions will disappear. Also calculate the rate of formation of sulphate ions.

Ans: $4.11 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$, $2.74 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$

Q8.

Q.3 For the following reaction



the rate of reaction is $\frac{d[\text{P}]}{dt} = k[\text{X}]$. Two moles of X are mixed with one mole of Y to make 1.0 L of solution. At 50 s, 0.5 mole of Y is left in the reaction mixture. The correct statement(s) about the reaction is(are)

(Use: $\ln 2 = 0.693$)

(A) The rate constant, k , of the reaction is $13.86 \times 10^{-4} \text{ s}^{-1}$.

(B) Half-life of X is 50 s.

(C) At 50 s, $-\frac{d[\text{X}]}{dt} = 13.86 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$.

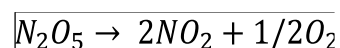
(D) At 100 s, $-\frac{d[\text{Y}]}{dt} = 3.46 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$.

Ans: B, C and D

Q9. The rate constant of a zero order reaction, $\text{A} \rightarrow \text{Products}$, is $2.1 \times 10^{-2} \text{ mol.L}^{-1} \text{ s}^{-1}$. If the initial concentration of A is 21 moles per litre, calculate its concentration after 15 minutes.

Ans: 2.1 mol L^{-1}

Q10. The initial concentration of N_2O_5 in the following first order reaction is $2.37 \times 10^{-3} \text{ mol.L}^{-1}$ at 298 K.



Calculate the rate constant for the reaction if the concentration of N_2O_5 is reduced to $0.38 \times 10^{-3} \text{ mol.L}^{-1}$ after 50 mins.

Ans: 0.031 min^{-1}

Q11. The rate constant for a first order reaction is $1.75 \times 10^{-2} \text{ s}^{-1}$ at 298 K. Calculate the time required to reduce the initial amount of reactant of 10 g to 4 g.

Ans: 52.37 s

Q12. For a chemical reaction: $A + B \rightarrow \text{Products}$, an experiment was performed with three different conditions. The observations for the same are tabulated as:

Exp.	Initial [A] (mol/L)	Initial [B] (mol/L)	Init. Rate of Formation of products (M s^{-1})
1	0.05	0.05	7.8×10^{-4}
2	0.10	0.05	1.56×10^{-3}
3	0.10	0.025	7.8×10^{-4}

Determine the rate law for this reaction and the rate constant. What will be the initial rate of reaction when $[A]_0 = 0.132 \text{ M}$ and $[B]_0 = 0.012 \text{ M}$?

Ans:

Q13. In a flask containing inert gas, PH_3 gas is introduced at 500°C . The decomposition of PH_3 gas to $\text{P}_4(g)$ and $\text{H}_2(g)$ happens with time. The total pressure is as a function of time is given in the table below.

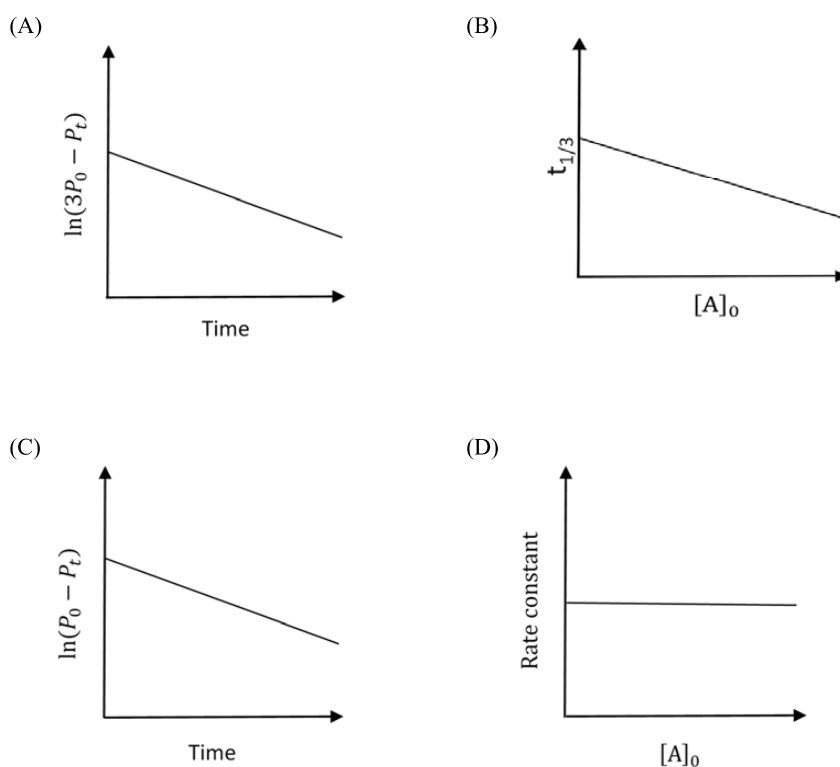
Time (sec)	0	60	120	∞
P (mm)	246.4	263.9	267.2	268.1

What is the order and rate constant of the reaction?

Ans: 1st order, 0.027 s^{-1}

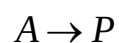
Q14.

- Q.5 For a first order reaction $A(g) \rightarrow 2B(g) + C(g)$ at constant volume and 300 K, the total pressure at the beginning ($t = 0$) and at time t are P_0 and P_t , respectively. Initially, only A is present with concentration $[A]_0$, and $t_{1/3}$ is the time required for the partial pressure of A to reach $1/3^{\text{rd}}$ of its initial value. The correct option(s) is (are)
(Assume that all these gases behave as ideal gases)



Ans: A and D

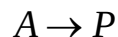
Q15. Derive the relationship between Y_0 , Y_∞ , Y_t , A_0 , t and rate constant k for a reaction given below, where reaction is of order $1/2$ w.r.t. A.



Ans:

$$2[A]_0^{1/2} \left[1 - \left(\frac{Y_t - Y_\infty}{Y_0 - Y_\infty} \right)^{1/2} \right] = kt$$

Q16. Derive the relationship between Y_0 , Y_∞ , Y_t , A_0 , t and rate constant k for a reaction given below, where reaction is of order 2 w.r.t. A.



$$\frac{Y_0 - Y_\infty}{Y_t - Y_\infty} = kt[A]_0 + 1$$

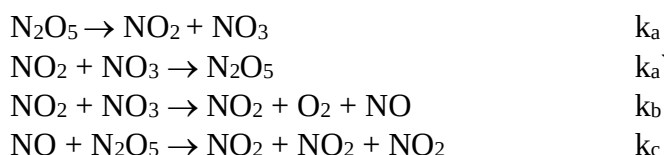
Q17. The reaction *cis*-2-butene to *trans*-2-butene is first order in both directions. At 25°C, the equilibrium constant is 0.406 and the forward rate constant is $4.21 \times 10^{-4} \text{ s}^{-1}$. Starting with a sample of pure *cis* isomer with $[\text{cis}]_0 = 0.115 \text{ mol.dm}^{-3}$, how long would it take for half the equilibrium amount of the *trans* isomer to form?

Ans: 490 s

Q18. Devise the rate law for the composition of N_2O_5 :



On the basis of the following mechanism:



Ans:

$$\left| \frac{d[\text{N}_2\text{O}_5]}{dt} \right| = \frac{2k_a k_b}{k_a' + k_h} [\text{N}_2\text{O}_5]$$

Q19. Plot the fraction present of each species versus the time elapsed for the following cases:

- (i) A in equilibrium with B (first order in both directions, $[A]_0 \neq 0$ and $[B]_0 = 0$), with $k_1/k_{-1} = 1$.
- (ii) A simultaneously decomposing into B and C with first-order rate constants k_1 and k_2 , respectively, and $k_2/k_1 = 3$ ($[A]_0 \neq 0$, $[B]_0 = 0$, and $[C]_0 = 0$).
- (iii) A in equilibrium with B and C simultaneously with first-order rate constants $\{k_1, k_{-1}\}$ and $\{k_2, k_{-2}\}$, respectively, and $k_1 = 1 \times 10^{-2}$, $k_{-1} = 1 \times 10^{-4}$, $k_2 = 1 \times 10^{-3}$, and $k_{-2} = 1 \times 10^{-7}$ (with proper units).
- (iv) A decaying to B and B decaying to C with first-order rate constants of k_1 and k_2 , respectively, and $k_1/k_2 = 20$ ($[A]_0 \neq 0$, $[B]_0 = 0$, and $[C]_0 = 0$).
- (v) A decaying to B and B decaying to C with first-order rate constants of k_1 and k_2 , respectively, and $k_2/k_1 = 20$ ($[A]_0 \neq 0$, $[B]_0 = 0$, and $[C]_0 = 0$).